CORE 200 Scientific Methods
Handout 27

Group name

Enter your details :

Section

Instructions/information: Following are some exercises that relate to a reading you have been assigned. Please refer to your notes as you answer them. You are encouraged to confer with your group mates as you put down your thoughts.

Reflect and discuss (A)

In each of the following scenarios, a conclusion is drawn about a population based on the sample results. Select which form of statistical inference this conclusion represents.

1. Based on a recent poll of 1,015 U.S. households that have an Internet connection, the researchers claimed with 95% confidence that among all U.S. households that have an Internet connection, between 63% and 67% have a high-speed link.

A. point estimation.
B. interval estimation.
C. hypothesis testing.

2. A company that provides coaching for the SAT claims in one of its ads: "90% of our students improve their SAT scores after attending our course." To check the company's claim, 500 students who took the company's course were sampled, and it was found that 407 of them (81.4%) actually improved their SAT scores after attending the course. Based on the sample results, we have some serious doubts regarding the accuracy of the company's claim.

A. point estimation.
B. interval estimation.
C. hypothesis testing.

3. In a recent study, 1,115 males 25 to 35 years of age were randomly chosen and asked about their exercise habits. Based on the study results, the researchers estimated that the mean time that a male 25 to 35 years of age spends exercising in a week is about 3.5 hours.

A. point estimation.
B. interval estimation.
C. hypothesis testing.

Reflect and discuss (B)

A study on exercise habits used a random sample of 2,540 college students (1,220 females and 1,320 males).

The study found the following:

- 818 of the females in the sample exercise on a regular basis.
- 924 of the males in the sample exercise on a regular basis.
- The average time that the 1742 students who exercise on a regular basis (818 + 924) spend exercising per week is 4.2 hours.

ctd. on next page

4. What is the point estimate for the proportion of all ____________ college students who exercise on a regular basis? (Numbers are rounded to two decimal places as needed.)

A. 0.32 B. 0.67 C. 0.69 D. 0.48

---

5. What proportion of the ____________ in the sample exercise on a regular basis?

---

6. Which of the following has a point estimate of ____________?

A. The mean time that all college students spend exercising per week.
B. The mean time that all college students who exercise on a regular basis spend exercising per week.
C. The percentage of all college students who exercise on a regular basis.

Reflect and discuss (C)

7. A researcher wanted to estimate μ, the mean number of hours that students at a large state university spend exercising per week. The researcher collects data from a sample of 150 students who leave the university gym following a workout.

Which of the following is true regarding ____________?

A. It is an unbiased estimate for μ.
B. It is not an unbiased estimate for μ and probably underestimates μ.
C. It is not an unbiased estimate for μ and probably overestimates μ.

Reflect and discuss (D)

8. Two opinion polls were conducted regarding the cultivation of Western ideals in academia. In particular, both polls were conducted in order to estimate p, the proportion of all Pakistani adults who believe that national educational institutions actively promote Western culture to the detriment of indigenous values.

1. The poll by the Gallow company was based on a random sample of 1,024 Pakistani adults and estimated the above proportion to be 0.60.
2. The New Research Center poll was based on a random sample of 2,002 Pakistani adults and estimated the above proportion to be 0.57.

Which of the two polls estimates p (the proportion of all Pakistani adults who believe that national educational institutions actively promote Western culture to the detriment of indigenous values) ____________?

A. The Gallow company poll.
B. The New Research Center poll.
C. Both polls estimate p with the same accuracy because both are based on a random sample.

Reflect and discuss (E)

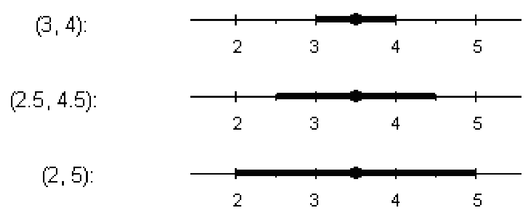
9. A study was done on pregnant women who smoked during their pregnancies. In particular, the researchers wanted to study the effect that smoking has on pregnancy length. A sample of 114 pregnant women who were smokers participated in the study and were followed until the birth of their child. At the end of the study, the collected data were analyzed and it was found that the average pregnancy length of the 114 women was 260 days. From a large body of research, it is known that length of human pregnancy has a standard deviation of 16 days.

ctd. on next page

Based on this study, find a ________ for μ, the mean pregnancy length of women who smoke during their pregnancy, and interpret your interval in context.

Reflect and discuss (F)

In a recent study, 1,115 males 25 to 35 years of age were randomly chosen and asked about their exercise habits. Based on the study results, the researchers estimated the mean time that a male 25 to 35 years of age spends exercising with 90%, 95%, and 99% confidence intervals. These were (not necessarily in the same order):



10. For which of the three intervals do you have the ________ *confidence* that it captures the population mean (μ)?

A. (3, 4).
B. (2.5, 4.5).
C. (2, 5).

11. The confidence interval in which you have the ________ confidence that it captures the population mean μ must be:

A. the 90% confidence interval.
B. the 95% confidence interval.
C. the 99% confidence interval.

12. Which of the three confidence intervals provides the ________ precise estimation?

A. (3, 4).
B. (2, 5).
C. (2.5, 4.5).

Reflect and discuss (G)

13. Suppose that based on a random sample, a 95% confidence interval for the mean hours slept (per day) among graduate students was found to be (6.5, 6.9). What is the ________ of this confidence interval?

A. 6.7.
B. 0.4.
C. 0.2.

Reflect and discuss (H)

14. A medical researcher wanted to estimate μ, the mean weight of newborns born to women over the age of 40.

The researcher chose a random sample of 100 pregnant women who were over 40, followed them through the pregnancy, and found that the mean weight of the 100 newborns was 3,035 grams. From past research, it is assumed that the weight of newborns has a standard deviation of $\sigma = 500$. The researcher calculated the 95% confidence interval for μ to be (2935, 3135).

If the researcher wanted to maintain the 95% level of confidence but report a confidence interval with a smaller ____________, which of the following will achieve that?

A. Redo the study, but this time with a sample of size 64.
B. Redo the study, but this time with a sample of size 225.
C. Redo the study and choose a different sample of size 100.

Reflect and discuss (I)

15. Below are four different situations in which a ____________ for μ is called for.

Situation A: In order to estimate μ, the mean annual salary of high-school teachers in a certain state, a random sample of 150 teachers was chosen and their average salary was found to be $38,450. From past experience, it is known that teachers' salaries have a standard deviation of $5,000.

Situation B: A medical researcher wanted to estimate μ, the mean recovery time from open-heart surgery for males between the ages of 50 and 60. The researcher followed the next 15 male patients in this age group who underwent open-heart surgery in his medical institute through their recovery period. (Comment: Even though the sample was not strictly random, there is no reason to believe that the sample of "the next 15 patients" introduces any bias, so it is as good as a random sample). The mean recovery time of the 15 patients was 26 days. From the large body of research that was done in this area, it is assumed that recovery times from open-heart surgery have a standard deviation of 3 days.

Situation C: In order to estimate μ, the mean score on the quantitative reasoning part of the GRE (Graduate Record Examination) of all MBA students, a random sample of 1,200 MBA students was chosen, and their scores were recorded. The sample mean was found to be 590. It is known that the quantitative reasoning scores on the GRE vary normally with a standard deviation of 150.

Situation D: A psychologist wanted to estimate μ, the mean time it takes 6-year-old children diagnosed with Down's Syndrome to complete a certain cognitive task. A random sample of 12 children was chosen and their times were recorded. The average time it took the 12 children to complete the task was 7.5 minutes. From past experience with similar tasks, the time is known to vary normally with a standard deviation of 1.3 minutes.

In which situation can we not use the confidence interval that we developed?

Reflect and discuss (J)

16. In order to estimate μ, the mean weight of a typical passenger + his/her luggage on an early morning flight from New York City to Washington DC, an airline company chose a random sample of 16 passengers who took the flight. From past research on similar flights, it is known that weights of passengers + their luggage have a normal distribution with standard deviation $\sigma = 20$ pounds.

Can we safely estimate μ with the formula for confidence interval that we developed?

A. Yes   B. No

Reflect and discuss (K)

17. For many years "working full-time" has meant 40 hours per week. Nowadays it seems that corporate employers expect their employees to work more than this amount. A researcher decides to investigate this hypothesis.

ctd. on next page

- *Claim 1:* The average time full-time corporate employees work per week is 40 hours.
- *Claim 2:* The average time full-time corporate employees work per week is ________ hours.

To substantiate his claim, the researcher randomly selects 250 corporate employees and finds that they work an average of 47 hours per week with a standard deviation of 3.2 hours.

In order to assess the evidence, we need to ask:

A. how likely it is in a sample of 250 we will find that the mean number of hours per week corporate employees work is as high as 47.
B. how likely it is that the true mean number of hours per week corporate employees work is 40.
C. how likely it is that the true mean number of hours per week corporate employees work is more than 40.
D. how likely it is that in a sample of 250 we will find that the mean number of hours per week corporate employees work is as high as 47 if the true mean is 40.

Reflect and discuss (L)

18. According to the Centers for Disease Control and Prevention, the proportion of U.S. adults age 25 or older who smoke is 0.22. A researcher suspects that the rate is lower among U.S. adults 25 or older who have a bachelor's degree or higher education level.

What is the ________ in this case?

A. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is less than 0.22.
B. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is 0.22.
C. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is not 0.22.
D. There is no relationship between education level and smoking habits.

---

19. What is the ________ in this case?

A. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is less than 0.22.
B. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is 0.22.
C. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is not 0.22.
D. There is a relationship between level of education level and smoking habits.

Reflect and discuss (M)

The following two hypotheses are tested:

- H_0: The proportion of Pakistani adults who oppose coeducation is roughly 50%.
- H_a: The proportion of Pakistani adults who oppose coeducation is above 50% (i.e., the majority oppose).

Suppose a survey was conducted in which a random sample of 1,100 Pakistani adults was asked about their opinions about coeducation, and based on the data, the p-value was found to be 0.002.

Comment: Throughout this question use a 0.05 (5%) significance level (cutoff).

---

20. The fact that the ________ means that:

ctd. on next page

A. There is 0.002 probability of observing data like those observed.
B. There is 0.002 probability that 50% of Pakistani adults oppose coeducation.
C. There is a probability of 0.002 (i.e., very unlikely) to observe data like those observed if the proportion of Pakistani adults who oppose coeducation were 50%.
D. There is 0.998 probability that the majority of Pakistani adults oppose coeducation.

---

21. Based on the __________ you can conclude that:

A. the data provide significant evidence that the proportion of Pakistani adults who oppose coeducation is 50%.
B. the data provide significant evidence that the majority of Pakistani adults oppose coeducation.
C. the data do not provide enough evidence to conclude that the majority of Pakistani adults oppose coeducation.
D. the data provide evidence that H_a is more likely than H_0 (i.e., it is more likely that the majority of Pakistani adults oppose coeducation).

---

22. Say that the __________ was not given, but rather, the following conclusion was advertised: “The survey does not provide enough evidence to conclude that the majority of Pakistani adults oppose coeducation.” Which of the following could have been the p-value that led to this conclusion?

A. 0.125.
B. 1.96.
C. 0.045.
D. --1.96.

---

23. When would you conclude that the data provide enough evidence that the proportion of Pakistani adults who oppose coeducation is __________?

A. when the p-value is small (less than 0.05).
B. when the p-value is not small (above 0.05).
C. when exactly half the individuals in the sample oppose coeducation and half support it.
D. never.

Reflect and discuss (N)

24. An NBC News/*Wall Street Journal* poll conducted in 2010 determined that 61 percent of Americans did not support the Tea Party movement. In a poll of 1,000 Americans this year, 64 percent say they do not support the Tea Party movement. Has opposition to the Tea Party movement increased since 2010?

We tested the following hypotheses at the 5 percent level of significance:

- H_0: The proportion of Americans this year who oppose the Tea Party movement is 0.61.
- H_a: The proportion of Americans this year who oppose the Tea Party movement is greater than 0.61.

The p-value is 0.026, so we reject the null hypothesis, H_0 , and accept the alternative hypothesis, H_a. We conclude that public opposition to the Tea Party movement is greater than 61% this year.

What type of error is possible here?

A. Type I.
B. Type II.

ctd. on next page

C. neither.
D. both.

Reflect and discuss (O)

25. A double-blind experiment is conducted to investigate the side effects of hormone replacement therapy for women with menopausal symptoms. The experiment randomly assigns more than 16,000 American women to either a hormone treatment or a placebo. After five years, the HRT study finds no significant difference in the proportion of women developing breast cancer and heart disease. Researchers decide, based on this finding, to allow the study to continue. As the null hypothesis was not rejected, there is a chance that the researchers made a type II error.

Given the type of error made in this situation, what could researchers do to ____________?

---

26. Suppose that at the end of the five-year study described above, a greater proportion of the hormone-treated group have breast cancer and heart disease. This observed difference is statistically significant. Researchers are so alarmed by the results that the experiment is ended early to prevent further harm to the health of the women participating in the hormone group. Since the null hypothesis was rejected, it is possible researchers made a type I error.

Given the type of error made in this situation, what could researchers do to ____________?